

The effects of microplastics and nanoplastics on cardiovascular disease: mechanisms and perspectives

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Abstract

Microplastics and nanoplastics (MNPs) are pervasive environmental pollutants that result from the degradation of plastic materials and the use of plastic-containing products. Although the accumulation of MNPs in the gastrointestinal and respiratory systems is well documented, growing evidence suggests that MNPs can translocate into the bloodstream and accumulate in cardiovascular tissue, raising concerns about their potential role in the development of cardiovascular disease. In this Review, we synthesize the current knowledge on MNP exposure routes, tissue distribution and biological effects relevant to cardiovascular health. We examine the latest clinical and experimental studies on the presence of MNPs in the blood, atherosclerotic plaques, thrombi and myocardial tissue, and critically evaluate the mechanistic evidence linking MNPs to endothelial dysfunction, atherosclerosis progression, myocardial injury and arrhythmogenesis. In vitro and in vivo data highlight plausible pathophysiological pathways linking MNPs to cardiovascular disease, including oxidative stress, mitochondrial dysfunction, inflammation and fibrotic remodelling. However, causal associations remain unproven in humans, and major methodological challenges persist, including inconsistent detection methods, limited epidemiological data and inadequate modelling of real-world exposure. We conclude by outlining research priorities and proposing a framework for how to integrate MNPs into environmental cardiology practice. As the global issue of plastic pollution intensifies, elucidating the cardiovascular risks posed by MNPs represents an urgent interdisciplinary challenge with substantial public health implications.

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Key points

- Microplastics and nanoplastics (MNPs) are increasingly detected in human cardiovascular tissues, including blood, atherosclerotic plaques and myocardium, suggesting systemic distribution and potential involvement in cardiovascular disease pathogenesis.
- Experimental studies reveal that MNPs induce oxidative stress, inflammation, mitochondrial dysfunction and cell death in cardiac cells, potentially contributing to myocardial damage, fibrosis and maladaptive remodelling.
- Detection methods for MNPs in biological samples remain heterogeneous, with limitations in sensitivity and standardization leading to under-reporting, especially for nanoplastics.
- Clinical evidence links MNP accumulation in thrombi and atherosclerotic plaques with increased pro-thrombotic states, higher inflammatory cytokines and worsened cardiovascular outcomes in human populations.
- Future research should prioritize improved biomonitoring, harmonized detection protocols, translational models and epidemiological studies to clarify MNP-related cardiovascular risks and inform regulatory action.

Introduction

Global plastic production has expanded at an unprecedented pace, increasing from 1.5 million tonnes in 1950 to >460 million tonnes in 2019, with this value projected to reach 1,100 million tonnes by 2050¹ (Box 1). This rapid and sustained growth has contributed to the widespread dispersion of plastic debris across all major environmental compartments, including air, water and soil. Through environmental degradation or direct industrial use, larger plastics break down or are manufactured into microplastics and nanoplastics (MNPs), comprising particles that are <5 mm and <1 µm in size, respectively. MNPs are now ubiquitously present in the human environment, with exposure occurring primarily through the ingestion of contaminated food and water, inhalation of airborne particles and, to a lesser extent, dermal contact. Estimates suggest that individuals might ingest or inhale >120,000 MNP particles annually^{2,3}. Notably, MNPs have been detected in human tissues, including the blood, lungs, placenta, liver, kidneys, faeces and, more recently, in atherosclerotic plaques, thrombi and cardiac tissue^{2–4}. These findings have raised important concerns about the potential effects of MNP accumulation on cardiovascular health.

Numerous reviews have examined the cardiovascular implications of MNP exposure, focusing on emerging toxicological data and mechanistic insights^{3,5}. Building on these contributions, in this Review, we synthesize the available evidence on this rapidly expanding field, emphasizing histopathological observations and mechanistic pathways linking MNP exposure to processes such as atherosclerosis progression, myocardial remodelling and thrombosis. We begin by characterizing MNPs in terms of their physical and chemical properties, classification and environmental behaviour. We then describe major human exposure routes and summarize the evidence of systemic distribution. A major section is devoted to the detection of MNPs in cardiovascular tissues, including atherosclerotic plaques, thrombi and the myocardium, which is followed by a discussion of the potential

pathophysiological mechanisms supported by in vitro and in vivo data, focusing on endothelial dysfunction, mitochondrial impairment, oxidative stress, inflammation and tissue remodelling. Finally, we consider the broader implications of MNP exposure for cardiovascular health and disease, highlight major knowledge gaps and offer suggestions for future research priorities.

Properties and types of MNPs

MNPs are a heterogeneous group of synthetic polymer particles that differ widely in their physical, chemical and biological characteristics⁶. These features, such as origin, shape, size, polymer composition and surface chemistry, collectively determine the environmental behaviour, bioavailability and toxicological potential of the MNPs⁷ (Fig. 1 and Table 1). An in-depth understanding of these characteristics is essential to assess their biological interactions and the mechanisms by which they might contribute to human disease, including cardiovascular pathology.

Classification by origin, shape, size and physicochemical properties

MNPs are commonly classified on the basis of their origin into two main categories: primary MNPs, which are intentionally manufactured at the microscale or nanoscale (such as cosmetic microbeads, industrial abrasives or medical carriers), and secondary MNPs, which arise from the degradation of larger plastic items through physical, chemical or biological processes⁷. The key contributors to the formation of secondary MNPs include fragmentation by ultraviolet radiation, thermal stress mechanical abrasion and microbial activity.

In terms of morphology, MNPs come in a wide range of shapes, including irregular fragments, cylindrical or spherical beads, fibres, foams, films, and granules with broken or jagged edges⁸. The shape of the MNP influences its biological interactions. For instance, compared with non-fibrous MNPs such as fragments, films or spheres, fibrous MNPs are more likely to accumulate at mucosal surfaces in the respiratory or gastrointestinal tracts, where they can induce chronic mechanical irritation and sustained inflammatory responses⁹. Conversely, fragmented particles with angular edges can damage epithelial membranes and are associated with increased cellular toxicity¹⁰.

Size is a defining factor in MNP toxicity. MNP particles are often categorized into large microplastics (1–5 mm), small microplastics (1 µm–1 mm) and nanoplastics (<1 µm). As particle size decreases, the surface-to-volume ratio increases, increasing the potential for adsorption of environmental pollutants, protein corona formation and cellular uptake¹¹. Nanoplastics, owing to their ultrasmall dimensions, can cross biological barriers such as the intestinal epithelium, blood–brain barrier and placenta, enabling systemic dissemination and intracellular interactions.

In addition to size and shape, other physicochemical properties (most notably surface charge, hydrophobicity and pigmentation) can substantially influence the fate and behaviour of MNPs in biological systems^{2,7,8}. For instance, hydrophobic particles can more readily interact with lipid membranes and accumulate within tissues, whereas surface charge can modulate protein binding and cellular uptake mechanisms.

Polymer composition and environmental degradability and toxicity

The polymeric composition of MNPs is closely tied to their industrial origin and intended use¹². The most frequently encountered polymers

in environmental and biological samples include polyethersulfone, polyethylene, polyethylene terephthalate (PET), polymethylmethacrylate (PMMA), polypropylene (PP), polystyrene (PS), polyurethane, polyvinyl alcohol and polyvinyl chloride (PVC)⁸. These materials are pervasive in consumer products and industrial applications owing to their structural versatility, chemical resistance and low manufacturing costs. However, the same properties that make plastics durable also confer resistance to environmental and biological degradation¹³. When released into the environment, macroplastics degrade into MNPs primarily through abiotic processes such as exposure to ultraviolet radiation, photooxidation, mechanical abrasion and thermal stress¹⁴. Biotic degradation, which involves microbial enzymatic activity, occurs at a considerably slower rate and is often incomplete. As a result, these polymers accumulate over time in ecological and biological systems¹⁵.

Importantly, degradation does not eliminate toxicity. By contrast, degradation might increase biological risk by releasing toxic monomers, residual solvents and a complex mixture of chemical additives incorporated during manufacturing into the environment^{16,17}. These degradation by-products, together with the parent polymers themselves, contribute to the overall toxicological profile of MNPs.

Emerging evidence suggests that toxicity is not uniform across polymer types, with PVC consistently showing greater cytotoxic potential than polyethylene or polypropylene, even when particle size and dose are controlled^{18,19}. Multiple *in vitro* studies, spanning epithelial, immune and advanced gut models, report that PVC induces stronger oxidative stress, greater depletion of antioxidant defences and more pronounced mitochondrial apoptosis than polyethylene or polypropylene²⁰. PVC promotes the generation of intracellular reactive oxygen species, the activation of IL-1 β , IL-6, tumour necrosis factor (TNF) and nuclear factor- κ B (NF- κ B) signalling, as well as apoptosis regulator BAX–cytochrome-*c*-mediated cell death at doses at which polyethylene and polypropylene remain largely inert¹⁹. These effects are attributed not only to the chloride-rich surface and leachable plasticizers in PVC, which are known to promote the formation of free radicals, but also to its greater affinity for pro-inflammatory protein coronas.

Together, these findings highlight the importance of considering polymer-specific toxicity in both environmental and health-risk assessments. Given that not all plastics behave equivalently after degradation or internalization, the chemical identity of MNPs is as important as their size, shape or concentration when evaluating their potential for harm.

Chemical additives and surface chemistry

Commercial plastics contain a complex mixture of chemical additives that are incorporated during manufacturing to modify their mechanical strength, flexibility, colour, flame resistance and ultraviolet stability. Among the most prevalent additives are plasticizers (such as phthalates), flame retardants (including polybrominated diphenyl ethers), stabilizers, colourants and ultraviolet absorbers. Many of these compounds are known to be toxic; for example, bisphenol A and some phthalates are well-characterized endocrine disruptors^{21,22}, whereas flame retardants such as polybrominated diphenyl ethers have been linked to neurotoxicity and carcinogenesis²³.

The surface chemistry of MNPs, including surface charge, functional groups and hydrophobicity, governs their environmental mobility, interaction with biological membranes and ability to adsorb co-contaminants^{8,24}. Aged or weathered MNPs that are exposed to ultraviolet light, oxidation or hydrolysis often exhibit increased surface

Box 1 | MNPs and global plastic pollution

Definitions

- Microplastics are plastic particles ranging from 1 μ m to 5 mm in size, whereas nanoplastics are defined as particles smaller than 1 μ m.
- Microplastics are further categorized into primary (intentionally manufactured at small size, for example, microbeads) and secondary (resulting from the breakdown of larger plastic debris through environmental degradation processes).

Global problem of plastic pollution

- More than 460 million tonnes of plastic are produced annually, with projections exceeding 1,100 million tonnes by 2050.
- MNPs have been detected across air, water, soil and food chains, with estimated human intake exceeding 120,000 particles per year.
- Mounting evidence shows that microscopic plastic debris infiltrates even remote ecosystems, raising concern about its persistence and potential biological effects.

MNPs, microplastics and nanoplastics.

roughness and reactivity, which can enhance their adsorptive affinity for proteins, lipids and xenobiotics^{25,26}.

Colourants are an often-overlooked chemical additive that might also have a role in MNP behaviour. Pigmentation can influence photooxidative degradation rates and biological responses²⁵. For instance, textile microfibres treated with azobenzene-based dyes have increased cytotoxicity and impaired epithelial integrity compared with their undyed counterparts²⁷. In addition, surface colour can affect microbial colonization patterns, potentially altering biofilm composition and toxicity^{25,28}.

Vectors for contaminants and microbial communities

Beyond their intrinsic properties, MNPs act as vectors for environmental contaminants and microbial communities^{29–31}. Their high surface area-to-volume ratio and chemical affinity for hydrophobic compounds allow them to adsorb a wide range of toxicants, including heavy metals (such as cadmium and lead), persistent organic pollutants (including dioxins and polychlorinated biphenyls) and polycyclic aromatic hydrocarbons^{29,30}. These interactions might result in co-exposure scenarios whereby the combined toxicity of the compounds exceeds that of the plastic particles or pollutants alone.

MNPs also function as substrates for biofilm formation, providing a niche for microbial colonization. These biofilms, collectively termed the ‘plastisphere’, might include bacteria, fungi, viruses and protozoa, some of which are pathogenic or carry antimicrobial-resistance genes³². The plastisphere facilitates horizontal gene transfer and can increase microbial virulence or persistence in biological systems^{32,33}. When internalized into the body through ingestion or inhalation, biofilm-coated MNPs might evade immune detection, persist longer than non-coated MNPs in tissues and contribute to systemic immune activation²⁸. Moreover, biofilms influence the aggregation, sedimentation and transport dynamics of MNPs in aquatic and biological environments³⁴. Their presence might alter the physicochemical behaviour of the particles, further complicating their clearance from biological systems and elevating the risk of chronic inflammation or infection.

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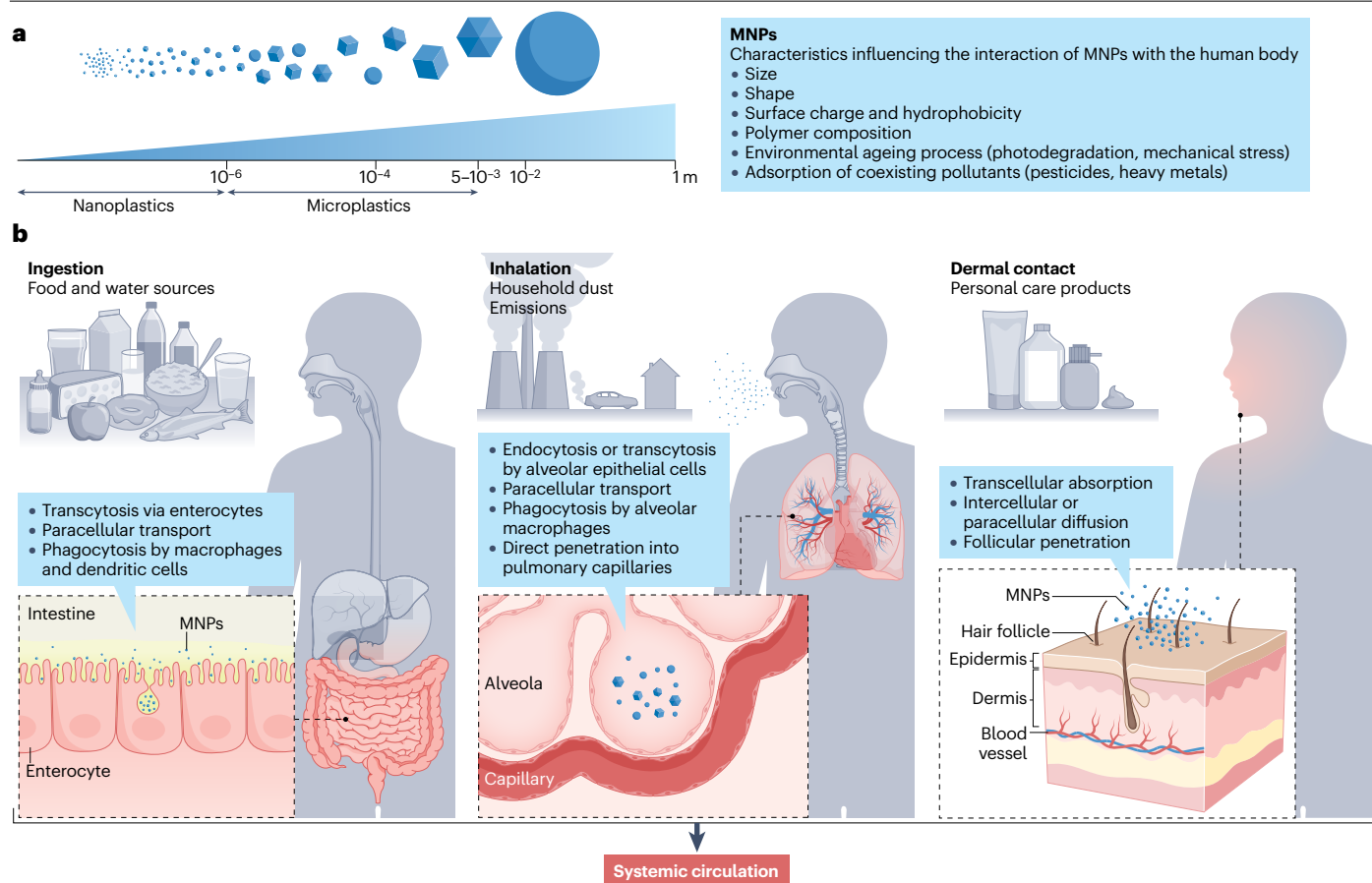


Fig. 1 | Routes and distribution of MNPs in the human body. a, The size spectrum of microplastics and nanoplastics (MNPs), ranging from nanometres (10⁻⁶ m) to millimetres (10⁻² m), with various characteristics influencing their interaction with the human body, including size, shape, surface charge, hydrophobicity, polymer composition, environmental ageing and adsorption of pollutants. **b,** The three primary exposure routes: ingestion, inhalation and dermal contact. Ingestion occurs through contaminated food and water, with MNPs crossing the gastrointestinal barrier via transcytosis, paracellular

transport or immune cell-mediated uptake. Inhalation involves exposure to MNPs present in household dust or emissions, which can penetrate the respiratory system through various mechanisms, including endocytosis, phagocytosis and direct entry into pulmonary capillaries. Dermal contact occurs via personal care products, with MNPs entering through transcellular absorption, intercellular diffusion or follicular penetration. These pathways can ultimately lead to MNPs reaching the systemic circulation, potentially causing adverse health effects.

Analytical detection and quantification methods

A comprehensive assessment of the health risks associated with MNPs depends on the ability to detect, characterize and quantify these particles in environmental and biological matrices. However, the current analytical landscape is fragmented, with limited standardization across methodologies. Broadly, the available techniques fall into two categories: particle-based and mass-based approaches, each with their distinct advantages and limitations³⁵.

Particle-based assessment approaches

Particle-based methods allow for morphological characterization and spatial resolution, but often lack chemical specificity for complex samples, especially in the nanometre range³⁵. Optical microscopy, including fluorescence-based imaging, can be used to visualize microplastics as small as 1 µm in size. Scanning electron microscopy, particularly when coupled with energy-dispersive X-ray spectroscopy, provides high-resolution imaging and elemental composition³⁶. However, owing to

resolution constraints, neither method is suited to detecting and characterizing nanoplastics³⁶. By contrast, dynamic light scattering can be used to estimate the hydrodynamic diameter and size distribution of particles in suspension and is particularly suited to nanoplastics. Nonetheless, this approach does not provide compositional information and is sensitive to sample aggregation, limiting its utility in heterogeneous biological matrices. Vibrational spectroscopy techniques, including Fourier transform infrared (FTIR) spectroscopy and Raman spectroscopy, are used to identify compounds by analysing the unique vibrations of their chemical bonds³⁷. Micro-FTIR enables polymer identification down to approximately 10 µm in size and is non-destructive (meaning that the MNPs do not need to be destroyed to identify their polymer composition), but water interference and spectral overlap can compromise accuracy³⁷. With submicrometre spatial resolution, Raman spectroscopy is well suited for identifying nanoplastics, but its performance might be affected by background fluorescence. FTIR and Raman spectroscopy are often used in combination to improve detection confidence³⁷.

Mass-based assessment approaches

Mass-based assessment methods offer high sensitivity and chemical specificity but are generally destructive and cannot provide particle-level morphological information. Pyrolysis–gas chromatography–mass spectrometry (Py-GCMS) can thermally decompose polymers into characteristic fragments for quantitative and qualitative analysis, providing high sensitivity and specificity for particle assessment, but does not preserve particle morphology³⁸. Thermal extraction desorption–gas chromatography–mass spectrometry allows for the semiquantitative detection of polymer types in complex biological matrices without complete combustion, preserving volatile compounds and enabling the analysis of tissue and biofluid samples. Differential scanning calorimetry measures thermal transitions in polymers, such as melting and glass transition points, aiding in bulk polymer characterization. Although less specific than chromatographic techniques, differential scanning calorimetry can complement other methods in mixed or poorly characterized samples. Inductively coupled plasma mass spectrometry uses an inductively coupled plasma to ionize the sample, allowing the doping of MNPs with trace metals (such as palladium) to increase detection sensitivity. The cytometry by time-of-flight method extends the sensitivity of this approach to single-cell resolution, enabling quantification of MNP uptake and immunotoxicity at the cellular level³⁸.

Of note, standardized protocols for the isolation, identification and quantification of MNPs are lacking, particularly for biological samples. Particle-based methods provide valuable insight into morphology and distribution but are limited in the detection of smaller nanoplastics and often lack throughput. By contrast, mass-based techniques offer precise chemical identification and quantification, but are destructive and lack spatial resolution³⁹. Emerging hybrid and multiomics platforms, such as scanning electron microscopy–Raman spectroscopy or Py-GCMS combined with inductively coupled plasma mass spectrometry, show promise in overcoming these limitations. Greater integration, validation and harmonization of these methodologies are essential to improve exposure assessment, enable cross-study comparability and support mechanistic toxicological investigations into MNP-related health effects.

Routes of human exposure to MNPs

Human exposure to MNPs occurs primarily through three main pathways: ingestion, inhalation and dermal contact⁴⁰ (Fig. 1).

Ingestion

Ingestion is considered the most prominent route of human exposure to MNPs, primarily due to the widespread contamination of food, beverages and packaging materials⁴⁰. An estimated 39,000–52,000 plastic particles are ingested annually through the diet alone⁴¹. MNPs have been detected in bottled and tap water, beer, honey, table salt, and particularly in seafood such as molluscs and fish, which tend to bioaccumulate these particles⁴⁰. Additional exposure occurs through the use of plastic containers and packaging, especially when exposed to heat⁴². For example, microwaving or storing hot food in plastic containers can release substantial amounts of MNPs into consumables⁴³.

When ingested, MNPs primarily interact with the gastrointestinal tract, which is the first and most important interface. Although a proportion of ingested MNPs is excreted, as evidenced by their detection in human faeces⁴⁴, studies have shown that smaller particles, particularly nanoplastics, can cross the intestinal epithelium and gain access to the systemic circulation^{3,5}.

Three primary mechanisms facilitate the translocation of MNPs across the intestinal barrier: paracellular transport via disrupted tight junctions, transcytosis through enterocytes and endocytosis by M cells in Peyer’s patches (aggregated lymphoid nodules)⁵. Numerous in vivo and in vitro studies have highlighted the role of particle size in translocation efficiency. Nanoplastics as small as 100 nm have been shown to penetrate the intestinal barrier and accumulate in internal organs, including the liver, kidneys and even the heart⁴⁵. Furthermore, PS nanoplastics have been shown to accumulate in the intestines and are internalized by enterocytes and goblet-like cells⁴⁶. Exposure to nanoplastics can lead to the reduced expression of tight junction proteins (such as tight junction protein 1 (TJP1) and occludin), altered expression of mucins (such as mucin 13) and increased intestinal permeability, contributing to a ‘leaky gut’ phenotype⁴⁶. This disruption is mediated through changes in host microRNAs (including miR-501-3p and miR-700-5p), which are upregulated by nanoplastics and interfere with tight junction integrity by targeting the gene encoding TJP1 (ref. 46).

Table 1 | Properties and characteristics of MNPs

Polymer composition	Size classification	Typical morphology	Environmental persistence	Additives	Degradation products
Polyethylene	MNPs	Fragments, fibres, pellets	High	Antioxidants and plasticizers	Smaller polyethylene fragments
Polystyrene	MNPs	Beads, fragments	High	Styrene monomer and flame retardants	Brittle MNPs
Polypropylene	Microplastics	Fibres, fragments	High	Ultraviolet stabilizers	Cracking and fragmentation
Polyvinyl chloride	Microplastics	Fragments	Very high	Phthalates, stabilizers	Chlorine-based compounds
Polyethylene terephthalate	Microplastics	Fibres, films	Moderate to high	Antimony catalysts	Terephthalic acid
Polyethersulfone	MNPs	Fibres, films	Very high	Solvents, colourants	Bisphenol S, sulfone-based fragments
Polymethylmethacrylate	MNPs	Beads, fragments	Moderate	Ultraviolet stabilizers, plasticizers	Methacrylic acid, formaldehyde
Polyurethane	MNPs	Foams, fragments	High to very high	Flame retardants, blowing agents	Isocyanates, amines, aromatic compounds
Polyvinyl alcohol	MNPs	Films, fibres	Low to moderate	Plasticizers, surfactants	Acetic acid, small alcohols

MNPs, microplastics and nanoplastics.

Nanoplastics also affect the composition of extracellular vesicles from intestinal cells, further influencing gene expression involved in barrier function.

Beyond structural disruption, nanoplastics have also been shown to alter extracellular vesicle composition and promote gut dysbiosis. Shifts in gut microbial communities, such as elevated concentrations of Ruminococcaceae and an upregulation in internalization by Lachnospiraceae, can impair mucin secretion and compromise the mucus barrier⁴⁶. Additional studies show that even low-dose, chronic exposure to microplastics affects gut microbial viability and metabolism. For instance, 10–100 µg/ml of 1-µm-sized PS microplastics has been shown to impair probiotic strains and reduce the production of microbial indoles that are important for immune signalling and epithelial homeostasis⁴⁷. In an *in vivo* study in mice, daily oral exposure to PS microplastics (1–10 mg/kg) altered the abundance of gut microbial taxa such as Erysipelotrichales and Lactobacillales, suppressed the production of tryptophan-derived metabolites, and disrupted host–microorganism interactions that are known to regulate inflammation and cardiovascular processes⁴⁸. These changes in the gut microbial ecosystem can also influence immune signalling, nutrient absorption and systemic inflammation.

Inhalation

Inhalation is an important, but often underappreciated, route of human exposure to MNPs. MNPs are widely dispersed in both indoor and outdoor air, originating from sources such as urban dust, synthetic textiles, tyre and road wear, construction materials, industrial emissions and waste incineration^{49,50}. The most prevalent MNP morphologies detected in the airborne fraction are fibres and fragments. Inhalation rates are estimated to range from 26 to 132 MNP particles per day, although exposure can be substantially higher in densely populated urban centres or certain occupational settings⁵¹. Indoor environments, particularly those with synthetic carpeting, upholstered furniture and air-conditioning systems, might have high concentrations of airborne MNPs, especially in poorly ventilated spaces. When inhaled, MNPs can deposit along various regions of the respiratory tract, from the nasal cavity and trachea to the bronchioles and alveoli, depending on particle size and shape⁵². Although the mucociliary clearance system has a key role in removing larger particles (>5 µm), smaller particles, especially nanoplastics, can evade clearance mechanisms, penetrate deep into the alveolar spaces and potentially translocate into the systemic circulation⁵³. Evidence from animal models suggests that inhaled particles <1 µm in size can cross the alveolar epithelium and enter the bloodstream, contributing to their systemic distribution⁵⁴.

Inhalation of MNPs has been linked to a range of respiratory pathologies due to the increased generation of reactive oxygen species, epithelial damage, altered cytokine profiles, and fibrotic and chronic tissue remodelling in response to prolonged exposure^{55,56}. These effects mirror those observed in occupational exposure to synthetic textile fibres, whereby persistent inhalation has been associated with interstitial lung disease, bronchiolitis and airway obstruction⁵⁷. Inhalation of nanoplastics by animal models leads to alveolar thickening, macrophage infiltration and DNA damage^{58,59}.

MNPs can also act as carriers of airborne environmental pollutants, including polycyclic aromatic hydrocarbons, heavy metals and volatile organic compounds⁶⁰. This phenomenon (often referred to as the 'Trojan horse' effect) can amplify the toxicological consequences of inhaled particles by facilitating the co-delivery of otherwise less bioavailable or more transient contaminants into deep lung tissue or

the systemic circulation⁶¹. Moreover, the hydrophobic surfaces of many synthetic polymers can enhance the adsorption of airborne pollutants, increasing the potential for local inflammatory responses, oxidative damage and cytotoxicity⁶². These effects might be further exacerbated by ageing and the surface modification of MNPs in the environment, which can increase their reactivity and biological interactions⁶².

Dermal contact

Although relatively underexplored compared with ingestion and inhalation, dermal exposure is a plausible route for human contact with MNPs, particularly through personal care products (such as facial scrubs, exfoliants and toothpaste), contaminated water and airborne deposition on to the skin^{40,51}. Synthetic polymers have been widely used as abrasives and bulking agents in cosmetics, often in micro-sized or nano-sized forms, facilitating direct and repeated skin contact. The human skin functions as a robust barrier against most external agents and effectively blocks larger microplastic particles, but nanoplastics (<100 nm) might penetrate through appendageal pathways such as hair follicles and sweat glands, or through microlesions or damaged skin⁶³. Experimental models have demonstrated that PS nanoplastics can localize in deeper skin layers under conditions of barrier impairment⁶⁴, raising concerns about chronic, low-level dermal absorption, particularly in occupational settings or with frequent cosmetic use.

Emerging studies suggest that prolonged or repeated skin exposure to MNPs can contribute to oxidative stress, cutaneous inflammation and the disruption of skin homeostasis, including altered keratinocyte function and impaired barrier recovery⁵⁵. Moreover, MNPs can act as carriers for chemical additives (such as phthalates and parabens) and environmental toxins adsorbed on to their surfaces, potentially increasing percutaneous toxicity. Nanoplastics have also been shown to interact with skin microbiota, potentially influencing local immune responses and barrier function.

Systemic translocation and distribution

Regardless of the route of entry, MNPs have been shown to translocate into the bloodstream, enabling systemic circulation and tissue distribution³. When internalized, MNPs can compromise endothelial barrier integrity, facilitating further vascular entry and systemic spread. Their interaction with plasma proteins, lipids and immune cells can trigger oxidative stress, inflammation and immune activation, all of which are implicated in chronic disease processes^{65,66}.

In addition to environmental exposure, intravenous medical devices are a direct and under-recognized source of microplastic contamination. Studies have shown that medical tools, such as syringes, intravenous bags and tubing, can release microplastic fragments (predominantly polypropylene and polyethylene) directly into the bloodstream⁶⁷. The amount of released MNPs depends on factors such as device type, manufacturer, usage frequency, and the presence or absence of inline filters. Although this exposure route contributes a smaller absolute burden compared with dietary or inhalation pathways, its direct vascular access and limited elimination capacity make it particularly concerning.

When in the bloodstream, MNPs have the potential to accumulate in various organs⁶⁸. Studies from the past 7 years have reported the presence of particles in a growing number of human tissues and biological fluids in addition to the blood, including the lungs^{52,69}, colon⁷⁰, liver⁷¹, kidneys⁷², placenta^{66,73}, spleen⁷¹, urine⁷⁴, stool⁷⁵ and sputum⁷⁶. These findings provide compelling evidence for systemic exposure to MNPs. In some tissues, concentrations as high as 138.9 particles

per gram have been reported, with a range of polymers identified, including polyethylene, polypropylene, PS and PET⁷⁷. The presence of MNPs in human placenta and fetal tissues raises particular concern for the developmental and transgenerational effects of MNP exposure. Moreover, disease states might influence MNP burden. For example, patients with inflammatory bowel disease have been shown to excrete higher numbers of MNPs (up to 41.8 particles per gram of faeces) compared with healthy individuals⁷⁸, suggesting altered retention or uptake associated with compromised intestinal barriers.

In summary, the capacity of MNPs to access and persist in the human circulatory system and internal organs highlights the urgent need to better understand their toxicokinetics, organotropism and long-term health consequences. Systemic exposure is not merely incidental but might be a key contributor to chronic inflammation, immune dysregulation and disease development across multiple organ systems.

MNP accumulation in cardiovascular tissues

Growing evidence supports the presence and accumulation of MNPs in human cardiovascular tissues (Table 2), suggesting a potential role in cardiovascular pathology. Given their presence in the systemic circulation, MNPs can translocate into cardiovascular structures, including the vascular endothelium, atherosclerotic plaques, thrombi, pericardial membranes and myocardial tissue^{3,5}. Their ability to bypass endothelial barriers and interact with immune and structural cells in the cardiovascular system has been increasingly demonstrated through advanced imaging and analytical techniques. In a landmark study, MNPs were found in 24 out of 30 thrombi samples collected from patients with ischaemic stroke, myocardial infarction or deep vein thrombosis⁷⁹, with median concentrations of MNPs in the thrombi of 62 µg/g, 142 µg/g and 70 µg/g, respectively. Among 15 polymer types identified, polyethylene was the most predominant, comprising 54% of all particles, with an average particle size of 36 µm (ref. 79). These findings suggest substantial deposition of MNPs in thrombotic material, particularly in the setting of acute vascular events.

With respect to atherosclerotic disease, study investigators have assessed Py-GCMS-quantified MNPs in carotid and coronary artery plaques and compared them with non-atherosclerotic aortic

segments⁸⁰. PET was the most abundant polymer detected, followed by polyethylene, PVC and polyamide 66 (PA66). MNP concentrations were higher in atherosclerotic tissues, indicating preferential accumulation in regions of vascular injury, inflammation and remodelling⁸⁰. Further supporting this finding, a separate study examined carotid endarterectomy specimens from 257 patients and found polyethylene in 58% and PVC in 12% of atherosclerotic plaques⁸¹. Electron microscopy showed jagged MNP fragments embedded in macrophage clusters and extracellular debris. Importantly, MNP-positive plaques were associated with elevated expression of pro-inflammatory cytokines, including IL-1β, IL-6, IL-18 and TNF, as well as immune markers such as CD3 and CD68, alongside reduced collagen content, suggestive of a more vulnerable and inflammatory plaque phenotype⁸¹.

A 2023 study reported the detection of MNPs in the human heart⁸². Cardiac and pericardial tissues from 15 patients undergoing surgery were analysed, revealing 9 polymer types, including PET, PMMA and polyurethane. MNPs were localized in the myocardium, pericardium, epicardial adipose tissue and left atrial appendage, with particle sizes ranging from 20 µm to 469 µm (ref. 82). The study also compared preoperative and postoperative blood samples and reported changes in MNP composition and size distribution, raising concerns about iatrogenic introduction of MNPs during surgery via medical devices, tubing or airborne contamination in the operating room⁸².

Collectively, these findings confirm the widespread presence of MNPs in cardiovascular tissues and support their potential involvement in vascular inflammation, atherosclerotic progression and thrombus formation. The preferential accumulation in diseased or inflamed tissues, coupled with their ability to interact with immune cells and promote cytokine release, underscores their relevance in cardiovascular pathology. Moreover, the possibility of medical device-associated contamination highlights an urgent need for procedural safeguards and further investigation into the iatrogenic risk of MNP exposure.

Clinical correlates of cardiovascular accumulation of MNPs
Although the available evidence is insufficient to establish a causal relationship between MNPs and cardiovascular disease, numerous studies published in the past 3 years have reported associations between the

Table 2 | Detection of MNPs in human cardiovascular tissues

Sample types	Detection methods used	Polymer types identified	Concentrations	Clinical correlations	Ref.
Blood	Py-GCMS	PET, PE, PS	1.6 µg/g	NA	65
Arteries	Py-GCMS	PET, PE, PVC	118.7 µg/g	Possible correlation with atherosclerotic plaque burden	80
Pericardium, epicardial or pericardial adipose tissues, myocardium, left atrial appendage, venous blood samples	LDIR	PMMA, PS, PET, PP, PVC, PU, PE	NA	NA	82
Thrombus (cerebral, coronary, venous)	Py-GCMS, LDIR, SEM	PE, PVC	61.75 µg/g (IS), 141.80 µg/g (MI), 69.62 µg/g (DVT)	Possible correlation with the severity of IS, MI and DVT	79
Atherosclerotic plaques	FTIR spectroscopy, SEM	PE, PVC, PP, PS	PE 21.7±24.5 µg/mg, PVC 5.2±2.4 µg/mg	Higher risk of MI, IS or death from any cause	81
Coronary blood and thrombi	Py-GCMS	PS, PVC, PP, PET, PE	PS 2.47±6.38 mg/kg, PE 45.61±26.25 mg/kg, PVC 10.1±7.7 mg/kg	Increased risk of MACE	83

DVT, deep vein thrombosis; FTIR, Fourier transform infrared; IS, ischaemic stroke; LDIR, laser direct infrared spectroscopy; MACE, major adverse cardiac events; MI, myocardial infarction; MNPs, microplastics and nanoplastics; NA, not available; PE, polyethylene; PET, polyethylene terephthalate; PMMA, polymethylmethacrylate; PP, polypropylene; PS, polymerized styrene; PU, polyurethane; PVC, polyvinyl chloride; Py-GCMS, double-shot pyrolysis–gas chromatography–mass spectrometry; SEM, scanning electron microscopy.

presence of MNPs in cardiovascular tissues and indicators of disease severity and adverse outcomes. In a study examining thrombi samples from patients with ischaemic stroke, myocardial infarction or deep vein thrombosis, MNPs were detected in 80% of samples, with varying concentrations depending on the clinical condition⁷⁹. Patients with MNP-positive thrombi had higher D-dimer levels compared with patients with MNP-negative thrombi ($8.3 \pm 1.5 \mu\text{g/l}$ versus $6.6 \pm 0.5 \mu\text{g/l}$, $P < 0.001$), suggesting an association between the presence of MNPs and increased fibrinolytic activity. Moreover, a positive correlation was observed between MNP concentrations and thrombus burden, with an adjusted β -coefficient of 7.72 (95% CI 2.01–13.43, $P < 0.05$)⁷⁹. Although these observations do not demonstrate causality, they support the hypothesis that MNPs might be involved in modulating the local cellular environment in thrombotic lesions.

In the context of chronic vascular disease, a prospective study of 304 patients undergoing carotid endarterectomy reported that the presence of MNPs in atherosclerotic plaques was associated with an increased risk of a composite clinical outcome of myocardial infarction, stroke or death over a mean follow-up duration of 34 ± 7 months⁸¹. Patients with MNP-positive plaques had a hazard ratio of 4.53 (95% CI 2.00–10.27, $P < 0.001$) for adverse outcomes compared with individuals with MNP-negative plaques. Histological analyses showed plastic fragments embedded among immune cells, as well as increased expression of inflammatory markers (including IL-1 β , IL-6 and TNF) and reduced collagen content, features consistent with a more vulnerable plaque phenotype⁸¹.

Similarly, in a cohort of 142 patients undergoing coronary angiography for acute myocardial infarction, MNPs were detected in a high proportion of coronary artery samples: PVC in 95.4% of samples, polyethylene in 71.8%, PA66 in 61.8% and PS in 43.6% (ref. 83). Higher PVC levels were found in patients who later experienced major adverse cardiovascular events, and a 10-unit increase in PVC content was associated with a 1.4-fold increase in the risk of major adverse cardiovascular events, suggesting a potential dose-dependent association⁸³.

Importantly, evidence shows that MNPs exert dose-dependent cardiovascular toxicity. In an in vitro study, 80-nm-sized PS particles ($0.1 \mu\text{g/l}$) incubated with human induced pluripotent stem cell-derived cardiomyocytes (hiPS cell-CMs) impaired calcium handling and mitochondrial function, with more pronounced oxidative stress and apoptosis at higher doses⁸⁴. In an in vivo study, a low dose of PP (-12 ng/ml) induced cardiac hypertrophy in mice via fibroblast growth factor 21 signalling and the activation of MAPK and PI3K–AKT pathways⁸⁵. These effects occurred at exposure levels that equate to estimates of exposure in humans (0.1 – 5.0 mg/day), underscoring potential real-world relevance⁸⁶.

Taken together, these findings demonstrate consistent epidemiological and histopathological associations between MNP accumulation and markers of cardiovascular risk, such as thrombogenicity, inflammation and adverse outcomes. However, although biologically plausible mechanisms, such as immune activation, endothelial disruption and pro-thrombotic signalling, have been proposed, the available data are observational and cannot confirm a causal role. Further longitudinal and mechanistic studies are needed to clarify whether MNP accumulation contributes directly to cardiovascular pathology or provides a marker of exposure in individuals with pre-existing disease.

Potential mechanisms of cardiovascular injury

The pathophysiological effects of MNPs on the cardiovascular system is a rapidly evolving area of research. Whereas epidemiological evidence

remains limited and largely correlative, a growing body of experimental work, including both in vitro cellular models and in vivo animal studies, has identified multiple biological pathways through which MNPs might affect cardiovascular structure and function. These mechanisms include endothelial dysfunction, oxidative stress, mitochondrial injury, inflammation, cell death and tissue remodelling (Fig. 2 and Table 3). The effect of MNPs on cardiovascular health might vary depending on particle size, shape, surface chemistry, polymer type and co-exposure to other environmental toxicants.

Endothelial dysfunction and atherogenesis

A large number of studies have explored the effect of MNPs on the vascular endothelium, a key regulator of vascular tone and homeostasis. In an in vitro study, exposure of human endothelial cells to PS particles resulted in impaired migration, disrupted cytoskeletal organization and increased expression of adhesion molecules, such as intercellular adhesion molecule 1 and vascular cell adhesion protein 1 (ref. 87). These effects were accompanied by oxidative stress, as evidenced by increased production of intracellular reactive oxygen species and downregulation of the antioxidant enzyme haem oxygenase 1 (HO1), as well as the activation of heat shock proteins, a hallmark of cellular stress response⁸⁷.

In vivo studies using apolipoprotein E-deficient mice (an established model of atherosclerosis) demonstrated that chronic oral exposure to PS microplastics led to increased atherosclerotic plaque area, expanded necrotic core size and reduced collagen content, indicating plaque instability⁸⁸. These structural changes were paralleled by increases in vascular oxidative stress and an upregulation in NF- κ B and nuclear factor erythroid 2-related factor 2–HO1 pathways⁸⁸. Importantly, another mouse study assessing nanoplastics showed that MNP exposure dysregulated lipid metabolism, increased circulating triglyceride levels, and promoted macrophage foam cell formation through the upregulation of the scavenger receptor MARCO and accumulation of long-chain acylcarnitines, which synergized with MNPs to increase lipid uptake and foam cell transformation⁸⁹. Together, these findings indicate that MNPs might exacerbate atherogenesis by compromising endothelial integrity, amplifying inflammatory signalling and disturbing lipid handling, especially under conditions of metabolic stress.

Mitochondrial dysfunction and energetic failure

Cardiomyocytes are highly energy-dependent cells with large mitochondrial content. Several in vitro and in vivo studies suggest that MNPs can localize in mitochondria and disrupt their function⁹⁰ (Fig. 3). Experimental exposure of hiPS cell-CMs to PS nanoplastics resulted in mitochondrial membrane depolarization, cristae disorganization, reduced ATP production and increased generation of reactive oxygen species, triggering the activation of apoptotic signalling pathways^{90–92}.

In preclinical studies, dietary exposure to PS microplastics in chickens and carp has been associated with histopathological evidence of myocardial injury, including cardiomyocyte swelling, interstitial oedema and necrosis^{93,94}. Biochemical analyses confirmed mitochondrial dysfunction, with reductions in mitochondrial membrane potential and increases in lipid peroxidation and pro-apoptotic markers, such as caspase 3 and the ratio between the apoptosis regulators BAX and BCL2 (ref. 93). Moreover, mitochondrial damage might release mitochondrial DNA, acting as a damage-associated molecular pattern that activates the cGMP–AMP synthase–stimulator of interferon genes pathway⁹⁵, a mechanism linked to chronic inflammation and tissue remodelling in cardiovascular disease.

Oxidative stress and inflammation

Oxidative stress is a central feature of MNP toxicity. Particle-induced production of reactive oxygen species can occur through direct redox reactions at particle surfaces, especially for weathered or surface-functionalized plastics. In fish models, PS nanoplastics

triggered Toll-like receptor 4 activation, NADPH oxidase upregulation, lipid peroxidation and mitochondrial fragmentation in cardiac tissues⁹⁴.

The activation of Toll-like receptors and the NOD-, LRR- and pyrin domain-containing protein 3 (NLRP3) inflammasome leads to the

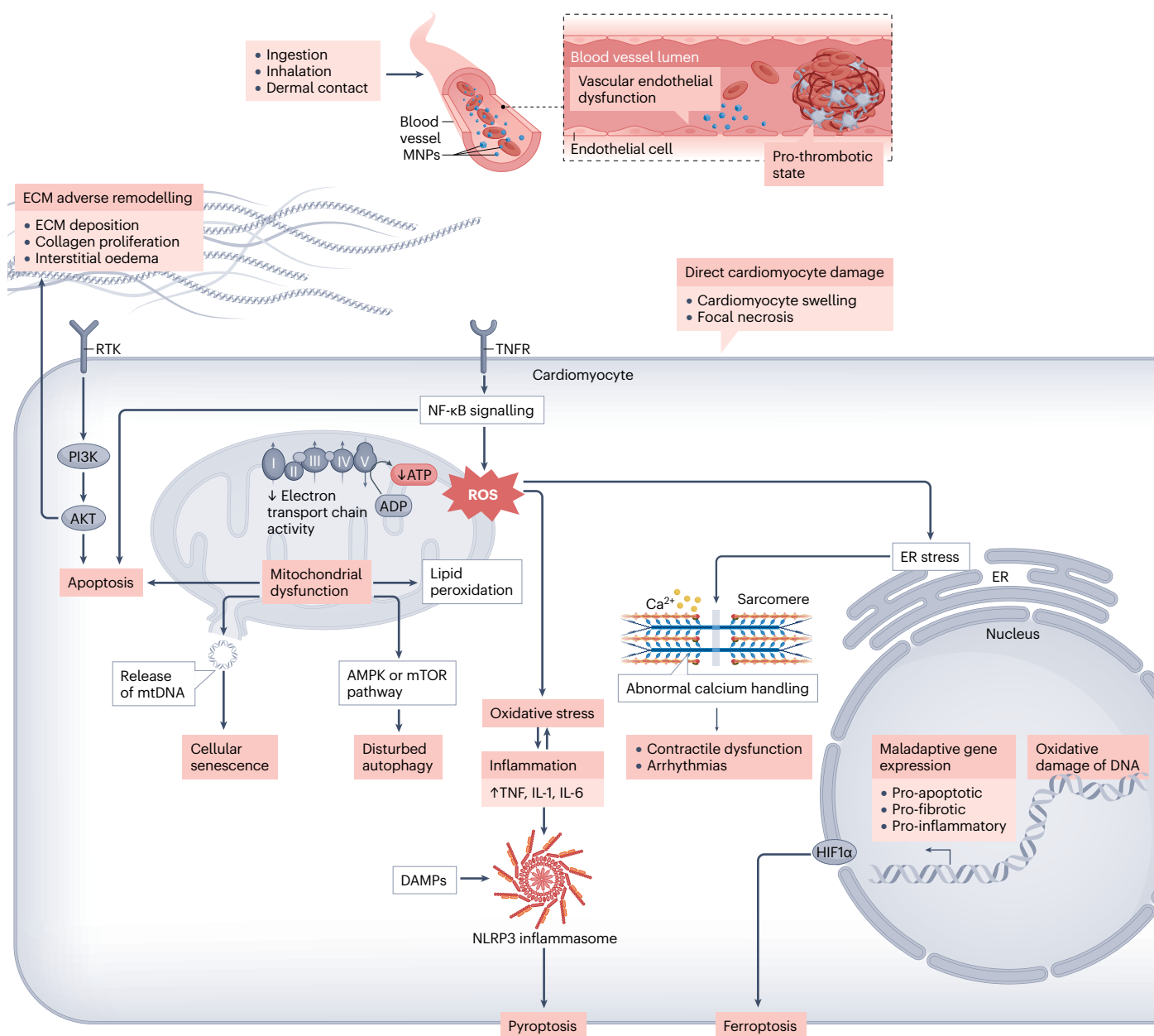


Fig. 2 | Cellular and molecular mechanisms of MNP-induced cardiovascular damage. The figure illustrates how microplastics and nanoplastics (MNPs) enter the systemic circulation through ingestion, inhalation and dermal contact, contributing to vascular endothelial dysfunction and a pro-thrombotic state. MNPs induce mitochondrial dysfunction through reactive oxygen species (ROS) generation, lipid peroxidation and reduced ATP production, resulting in oxidative stress, inflammation, apoptosis, autophagy disturbance and cellular senescence. The disruption of calcium handling by the endoplasmic reticulum (ER) and the resulting contractile dysfunction and arrhythmias are depicted. Additional

pathways include adverse extracellular matrix (ECM) remodelling, ferroptosis, pyroptosis and oxidative damage of DNA. These processes trigger maladaptive gene expression, promoting pro-apoptotic, pro-fibrotic and pro-inflammatory responses, ultimately contributing to cardiovascular damage. AKT, protein kinase B; AMPK, AMP-activated protein kinase; DAMPs, damage-associated molecular patterns; HIF1α, hypoxia-inducible factor 1α; mtDNA, mitochondrial DNA; mTOR, mechanistic target of rapamycin; NF-κB, nuclear factor-κB; NLRP3, NOD-, LRR- and pyrin domain-containing protein 3; PI3K, phosphoinositide 3-kinase; RTK, receptor tyrosine kinase; TNF, tumour necrosis factor; TNFR, tumour necrosis factor receptor.

Table 3 | Possible mechanisms underlying MNP-induced cardiovascular toxicity

Pathways involved	Key molecular targets	Downstream consequences
Endothelial dysfunction	ZO1, VE-cadherin, IL-6, TNF ^{87,88} , MARCO ⁸⁹	Pro-atherogenic environment (increased vascular permeability, endothelial injury, increased intracellular total cholesterol content and lipid droplet accumulation in macrophages)
Mitochondrial dysfunction	NF- κ B ⁹³ , NLRP3 inflammasome ⁹³ , caspase 1 (ref. 93), GPX4 (ref. 97), HO1, HIF1 α ⁹⁷ , γ H2AX, STING ⁹⁵	Cardiac systolic dysfunction, myocardial toxicity, myocardial fibrosis
Inflammation and oxidative stress	ROS ^{106,107} , caspase 3 (ref. 108), GSH-Px, SOD, CAT ¹⁰² , VEGF ¹⁰⁹ , CYP1A ¹¹⁰ , IL-1 (ref. 111)	Pericardial oedema, diastolic dysfunction, arrhythmias, bradycardia, left ventricular dilatation, vascular inflammation, cardiac malformations, thrombosis, angiogenesis inhibition
Apoptosis	Caspases, NLRP3, RIPKs, Nrf2, MAPK ^{98,112}	Fibrosis, myocardial fibre rupture, cardiac dysfunction, increased troponin T concentration, increased CK-MB activity
Fibrosis	AKT, BCL2, TGF β , β -catenin, METTL3, PI3K ^{100–102}	Cardiac hypertrophy
Sarcolemmal remodelling	TNNT2a ¹⁰³	Pericardial oedema, reduced heart rate, histopathological heart damage, vascular defects, thrombosis
Genomic modifications	m ⁶ A–ncRNA	Cardiac toxicity

AKT, protein kinase B; BCL2, apoptosis regulator BCL2; CAT, catalase; CK-MB, creatine kinase-MB; CYP1A, cytochrome P450 1A; GPX4, glutathione peroxidase 4; GSH-Px, glutathione peroxidase; HIF1 α , hypoxia-inducible factor 1 α ; HO1, haem oxygenase 1; MAPK, mitogen-activated protein kinase; METTL3, methyltransferase-like 3; m⁶A–ncRNA, N⁶-methyladenosine-modified non-coding RNA; NF- κ B, nuclear factor κ B; NLRP3, NOD-, LRR- and pyrin domain-containing protein 3; Nrf2, nuclear factor erythroid 2-related factor 2; PI3K, phosphoinositide 3-kinase; RIPKs, receptor-interacting protein kinases; ROS, reactive oxygen species; SOD, superoxide dismutase; STING, stimulator of interferon genes; TGF β , transforming growth factor- β ; TNNT2a, cardiac muscle troponin T type 2a; VEGF, vascular endothelial growth factor; γ H2AX, phosphorylated H2A histone family member X.

release of pro-inflammatory cytokines, such as IL-1 β and IL-18. These inflammatory pathways are also implicated in pyroptotic cell death, a highly inflammatory form of programmed cell death that contributes to myocardial injury. Transcriptomic analyses of MNP-exposed hearts identified upregulation of immune and inflammatory gene networks⁹⁶, suggesting chronic immune activation in response to plastic exposure.

Apoptosis, ferroptosis and other forms of cell death

Cardiomyocytes exposed to MNPs can result in multiple forms of cell death. Apoptosis is triggered by mitochondrial dysfunction and oxidative stress. Ferroptosis, a form of cell death driven by iron-dependent lipid peroxidation, has been demonstrated in cardiomyocytes exposed to PS nanoparticles⁹⁷. Furthermore, iron accumulation has also been shown to reduce expression of the phospholipid hydroperoxidase glutathione peroxidase 4 and activate the hypoxia-inducible factor 1 signalling pathway in cardiomyocytes, consistent with ferroptotic death⁹⁷. In severe cases, necrosis might occur due to irreparable

mitochondrial injury and energy depletion⁹⁸. Co-exposure studies (such as co-exposure of PS nanoplastics with cadmium) have shown simultaneous activation of pyroptosis, apoptosis and necrosis in cardiomyocytes, a phenomenon termed PANoptosis that has been shown to exacerbate cardiac inflammation and dysfunction⁹⁹.

Cardiac hypertrophy and fibrosis

Histological analyses of hearts from MNP-exposed rodents have revealed evidence of adverse remodelling, including cardiac hypertrophy and fibrosis¹⁰⁰. These changes are associated with increased expression of extracellular matrix components such as collagen I, collagen III and fibronectin, as well as dysregulation of signalling pathways, including PI3K–AKT–apoptosis regulator BCL2 (ref. 100) and Wnt– β -catenin¹⁰¹. In mouse models, exposure to airborne PS nanoplastics induced ventricular wall thickening, myocardial fibrosis and electrical conduction abnormalities¹⁰⁰. These effects might be the result of direct cardiomyocyte injury and/or fibroblast activation.

Calcium handling and electrophysiological disruption

Emerging studies show that MNPs can interfere with calcium homeostasis and ion channel function¹⁰². In hiPS cell-CMs and zebrafish embryos, PS nanoplastics disrupted sarcolemmal calcium channels, suppressed calcium ATPases, and altered the expression of genes crucial for contractility and electrical conduction, including *MYH6*, *TNNT2a* and *NKX2.5* (encoding myosin 6, cardiac muscle troponin T type 2a and homeobox protein Nkx-2.5, respectively)¹⁰³. These effects translated into electrophysiological changes, including prolonged action potentials, increased arrhythmogenicity and reduced cardiac output. Such disruptions could contribute to arrhythmia development, particularly under conditions of stress.

Epigenetic and transcriptomic modifications

Finally, MNP exposure might result in long-term changes in gene expression through epigenetic mechanisms. One study reported methylation changes in non-coding RNAs involved in endocytosis, cellular senescence and cell cycle signalling pathways in response to MNP exposure¹⁰⁴, implicating these regulatory elements in the response of cardiac tissue to plastics.

Overall, although causality cannot be definitively inferred from the available data, experimental evidence indicates that MNPs can engage multiple interrelated pathways associated with cardiovascular dysfunction, including oxidative stress, mitochondrial injury, inflammation, endothelial dysfunction, cell death and fibrotic remodelling. The presence of MNPs in human cardiovascular tissues, together with evidence of adverse remodelling in animal models, provides a compelling rationale for further mechanistic and epidemiological investigation into the cardiovascular consequences of MNP exposure.

Future directions

The detection of MNPs in human cardiovascular tissues, including blood, thrombi, atherosclerotic plaques and myocardium, has raised legitimate concerns about their potential involvement in the development of cardiovascular disease. A growing body of in vitro and in vivo research suggests that MNPs can disrupt key physiological processes by inducing endothelial dysfunction, oxidative stress, mitochondrial impairment and low-grade chronic inflammation. These biological effects have been linked to accelerated atherosclerosis, increased plaque instability and myocardial remodelling in experimental models. However, although these findings have compelling mechanistic

plausibility, they remain insufficient to establish causality in humans and should be interpreted with caution¹⁰⁵.

At present, the field faces substantial methodological and translational challenges. Many experimental studies use acute, high-dose exposures that do not accurately reflect real-world human exposure, which typically occurs chronically at lower doses through ingestion, inhalation and dermal contact. In vitro models often rely on monocultures or simplified systems that lack the physiological complexity of the cardiovascular environment, such as immune regulation, haemodynamic forces and comorbid conditions. Human epidemiological data remain limited in scope, with few large-scale or longitudinal studies evaluating MNP exposure in relation to cardiovascular outcomes. Moreover, important confounders such as diet, smoking, occupation, socioeconomic status and environmental co-exposures are rarely controlled for in a systematic manner, further limiting interpretability.

Another major barrier to progress is the absence of standardized methods for isolating, quantifying and characterizing MNPs in biological matrices, particularly in the nanometre size range. Although technologies such as Py-GCMS, Raman and FTIR spectroscopy, and laser direct infrared spectroscopy have proven useful, interstudy variability in sample processing, detection thresholds and reporting conventions impedes reproducibility and data harmonization. Standardized protocols and cross-validated benchmarks are urgently needed to enable consistent detection and accurate exposure assessment across different studies and laboratories.

The complexity of MNP toxicology is further amplified by their physicochemical heterogeneity. Polymer type, size, shape, surface charge, degree of weathering and adsorbed co-contaminants all modulate biological behaviour and toxicity. As stated above, given that certain polymers are associated with higher cytotoxicity than others, polymer-specific toxicity must be more rigorously addressed. In addition, MNPs frequently harbour hazardous additives (such as plasticizers and flame retardants) and environmental toxins (including heavy metals and polyaromatic hydrocarbons), which might synergize or confound the biological effects of the plastic particles themselves.

Disentangling these overlapping contributions remains an unmet need in both experimental and epidemiological research.

Similarly, the interaction between MNP exposure and conventional cardiovascular risk factors such as diabetes mellitus, obesity, hypertension and dyslipidaemia has not been adequately investigated, despite theoretical grounds for potential synergy. These conditions are characterized by systemic inflammation, metabolic dysfunction and compromised barrier integrity, which might increase MNP adsorption and toxicity. Future studies should aim to capture these interactions using integrated models and stratified population analyses.

Future research priorities should focus on refining exposure assessment tools and developing translational models that replicate chronic, low-level, environmentally relevant exposure. High-resolution analytical methods that can detect and characterize MNPs in diverse biological matrices are crucial. Experimental systems, including long-term animal studies and human-relevant in vitro models such as organoids and microfluidic chips, should be used to elucidate dose–response relationships and clarify mechanisms of cardiovascular injury. Large, well-powered cohort studies with longitudinal follow-up and rigorous adjustment for confounding factors are essential to determine the clinical relevance of MNP exposure. Furthermore, studies incorporating genomic, epigenomic and metabolomic profiling might identify early biomarkers of exposure and vulnerability, improving risk stratification.

From a public health perspective, MNPs should be considered emerging environmental exposures that might exacerbate cardiovascular risk, particularly in vulnerable populations. Integrating MNPs into the broader framework of environmental cardiology, alongside established pollutants such as fine particulate matter and endocrine disruptors, would foster a more comprehensive understanding of non-traditional risk factors for cardiovascular disease. Regulatory action to reduce environmental plastic contamination, through source reduction, safer polymer alternatives, improved waste management and restrictions on intentionally added microplastics, might ultimately contribute to improved cardiovascular and environmental health.

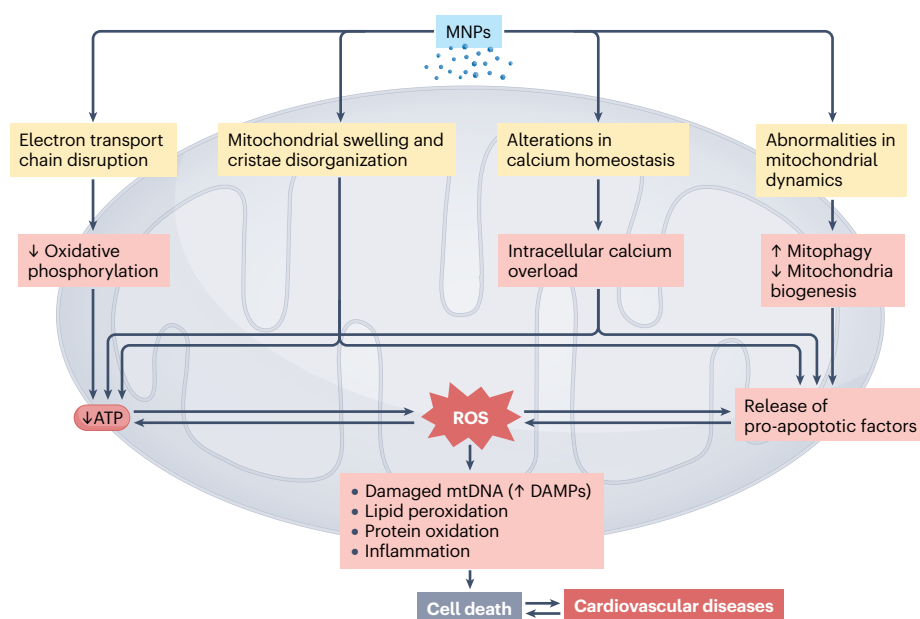


Fig. 3 | Mitochondrial dysfunction and cell death pathways in cardiovascular cells exposed to MNPs. The figure summarizes the mitochondrial alterations induced by microplastics and nanoplastics (MNPs) in cardiovascular cells. MNP exposure disrupts the electron transport chain and oxidative phosphorylation system, leading to decreased ATP production. Simultaneously, MNPs induce mitochondrial swelling and cristae disorganization, intracellular calcium overload and abnormalities in mitochondrial dynamics. Taken together, these effects lead to excessive production of reactive oxygen species (ROS) and the release of pro-apoptotic factors. Excessive ROS production damages mitochondrial DNA (mtDNA), induces lipid peroxidation and protein oxidation, and promotes inflammatory responses. These events converge on mitochondrial-driven cell death, ultimately contributing to the development of cardiovascular diseases. DAMPs, damage-associated molecular patterns.

Conclusion

Although definitive evidence of a causal relationship between MNP exposure and human cardiovascular disease is lacking, early experimental and clinical findings highlight the plausibility of such a link and support the need for rigorous investigation. As the global production of plastics continues to rise, understanding the cardiovascular implications of MNPs is an urgent and timely challenge. Addressing this challenge will require interdisciplinary collaboration, methodological standardization, and a shared recognition of the interconnectedness of environmental and human health.

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Author contributions

A.A., G.P. and E.T. researched data for the article and wrote the manuscript. A.A., E.R.-L., E.B. and A.B.-G. contributed to the discussion of content and reviewed and edited the manuscript before submission.

Competing interests

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